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ARTICLE

Scaling law of static liquefaction mechanism in geocentrifuge and corresponding hydromechanical characterization of an unsaturated silty sand having a viscous pore fluid

Amin Askarinejad, Alexander Beck, and Sarah M. Springman

Abstract: Fast landslides induced by rainfall impose considerable damage on infrastructure and cause major casualties worldwide. Static liquefaction is one of the triggering mechanisms mentioned frequently in the literature as a cause of this type of landslide. The scaling laws required to model this mechanism in the geotechnical centrifuge are developed, and it is shown that either a reduction in the soil pore size or use of a viscous pore fluid is needed to unify the time scaling factors of contractive volume change of the saturated voids and dissipation of the excess pore pressure generated. The latter option was used in this research; therefore, the influences of the viscous pore fluid on the hydromechanical characteristics of a silty sand were investigated. Subsequently, geocentrifuge tests were conducted to compare the behaviour of a slope having a viscous solution as the pore fluid with that of a model with water as the pore fluid. Both slopes were subjected to rainfall, and the evolution of the pore pressure and surface movements were monitored.

Key words: landslide, rainfall, static liquefaction, unsaturated soils, viscous pore fluid, geotechnical centrifuge modelling.

Résumé : Les glissements de terrain rapides provoqués par les précipitations pluvieuses infligent des dommages considérables aux infrastructures et font de nombreuses victimes dans le monde entier. La liquéfaction statique est l'un des mécanismes déclencheurs de ce type de glissement de terrain souvent évoqué dans la littérature. Les lois d'échelle nécessaires à la modélisation de ce mécanisme en centrifugeuse géotechnique sont élaborées. De plus, il est démontré qu'une réduction de la taille des pores du sol ou l'utilisation d'un fluide interstitiel visqueux est nécessaire à l'unification des échelles de temps de la variation du volume par contraction des espaces vides saturés et de la dissipation de l'excès de pression interstitielle généré. La dernière option a été utilisée dans la présente étude et l'influence du fluide interstitiel visqueux sur les caractéristiques hydromécaniques d'un sable silteux a donc été étudiée. Par la suite, des essais en centrifugeuse ont été effectués pour comparer le comportement d'une pente contenant une solution visqueuse en guise de fluide interstitiel à celui d'un modèle se servant d'eau comme fluide interstitiel. Les deux pentes ont été soumises à des précipitations pluvieuses et l'évolution de la pression interstitielle et des mouvements de surface ont été observés. [Traduit par la Rédaction]

Mots-clés : glissement de terrain, précipitation pluvieuse, liquéfaction statique, sols insaturés, fluide interstitiel visqueux, modélisation en centrifugeuse géotechnique.

Introduction

Landslides induced by long and (or) intense rainfall events affect lives and infrastructure severely in many parts of the world (e.g. Larsen et al. 2001; Lagmay et al. 2006; Cascini et al. 2008). These landslides are initiated by a decrease in the effective stresses, and hence the shear strength of the soil (e.g., Bishop and Morgenstern 1960; Campbell 1975; Brand 1981; Springman et al. 2013). Several attempts have been undertaken to investigate the triggering mechanisms of these frequent, and often powerful, natural hazards. One of the commonly proposed mechanisms of fast landslides triggered by rainfall is "static liquefaction", which is generally defined as the loss of strength of loose contractive soil under undrained conditions (Knill et al. 1976; Olson et al. 2000; Chu et al. 2003; Casini et al. 2010; Take and Beddoe 2014). Undrained shearing of soil elements with contractive behaviour may lead to effective stress paths with rapidly decreasing shear stresses, which nonetheless lie below the expected failure envelope. This is often described as "instability", although it is not synonymous with failure. Both instability and failure may lead to catastrophic events (Lade 1989; Eckersley 1990; Nova 2003; Take et al. 2004). Failure is defined as the state of the intersection of the stress path with the failure envelope.

Small disturbances in load may produce undrained conditions in loose soils with medium to low permeabilities, such as fine sands and silts, and large deformations may develop in the ground. When the instability condition is reached, the soil may or may not be able to sustain the current stress state (Nova 2003). This stress state corresponds to the maximum current effective stress (points D, E, and F in Fig. 1) on the path (ESP, for example, AD, BE, CF), as shown in Fig. 1 schematically in the p'–q space (where $p' = (\sigma_1' + \sigma_2' + \sigma_3')/3$; $q = \sigma_1 - \sigma_3$; p' is the mean effective stress; q is the deviatoric stress; σ_1' is the major principal effective stress; σ_2' is the intermediate principal effective stress; σ_3' is the minor principal effective stress; σ_1 is the major principal stress; and σ_3 is the minor principal stress). Following this peak point, unlimited plastic

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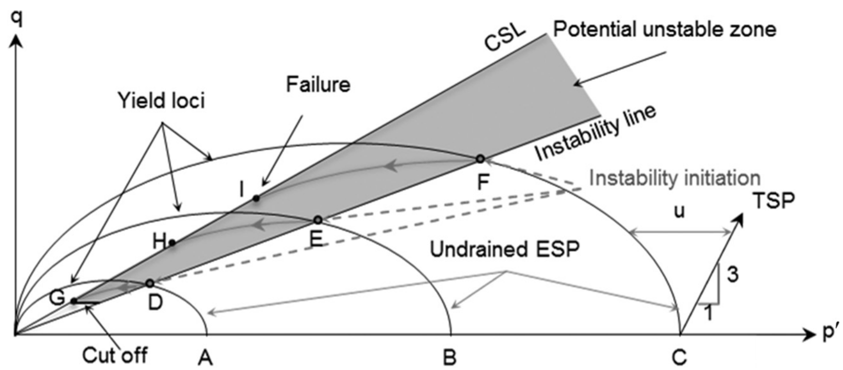
Fig. 1. Location of instability line and potential unstable region for undrained saturated triaxial tests (ESP, effective stress path; TSP, total stress path; CSL, critical state line) (Askarinejad 2013, after Chu et al. 2003).



strains may occur under decreasing stresses. The shaded region is bounded by the critical state line (the line passing through points G, H, and I) and the instability line (the line passing through points D, E, and F). It does not, however, extend down to the origin of stress space, but terminates by an approximately horizontal cut-off at fairly constant q (Lade 1993; Sasitharan et al. 1993; Vaid and Eliadorani 1998). The value of q at the cut-off line depends on the initial placement density of the soil. The position of this cut-off line relative to the origin of stress space moves to increasing q as initial density increases (Vaid and Chern 1985).

It has been shown that there is no intrinsic instability line, and its location is dependent on the rate of volume change or pore-pressure buildup (burst) (Take et al. 2004). Other studies showed that the locations of the instability lines are also strongly dependent on the void ratio (e.g., Chu et al. 2003). Di Prisco et al. (1995) noted that the instability line for undrained tests is also a function of stress path.

Drained collapse behaviour is a process that is similar to static liquefaction and can accelerate the instability in "drained" and undrained conditions. The buildup of pore-water pressure is directly related to the volume change characteristics of the soil (Lade and Yamamuro 2011). This contraction then results in development of pore-water pressures and associated reduction in shear resistance.

Centrifuge modelling of rainfall-induced landslides with controlled material properties, and boundary conditions, has the potential to provide an understanding of the triggering mechanisms of landslides due to rainfall. Several failure mechanisms of rainfall-induced landslides have been investigated by different researchers using centrifuge modelling. Take et al. (2014) demonstrated that the velocity and distal reach of a landslide induced by rainfall is strongly affected by the antecedent groundwater conditions. Similarly, Kimura et al. (1991) observed that fills with higher initial void ratios failed at lower intensities of rainfall, and when void ratios were similar, more rainfall was necessary to cause failure in the fills with lower initial water content.

However, very few evidences (e.g., Take and Beddoe 2014) of static liquefaction have been observed in tests using the geotechnical centrifuge modelling technique, although this mechanism has been frequently mentioned in the literature as the triggering mechanism of fast landslides. Take et al. (2004) performed a series of centrifuge tests on slopes of decomposed granite subjected to heavy rainfall. This material has shown potential for static liquefaction, based on the undrained triaxial tests on very loose and saturated samples. The centrifuge models exhibited significant losses of suction and collapse of the voids due to wetting, but neither buildup of positive pore pressure nor failure was observed. Take et al. (2004) concluded that the unsaturated condition of the soil (as pointed out also by Kimura et al. 1991) and different scaling laws for the dissipation rate of pore pressure and falling velocity of a particle at increased gravitational acceleration might have been the reasons that no failure was triggered in this slope by the static liquefaction mechanism.

Accordingly, the scaling laws for centrifuge modelling of the collapse of the voids in an unsaturated media and the dissipation rate of the possible resulting increase in pore-water pressure are studied in this paper. A series of centrifuge tests were performed using the geotechnical drum centrifuge at ETH Zurich, which was equipped with a rain simulator (Askarinejad et al. 2012*b*) to compare the responses of unsaturated loose silty sand models with water and a viscous solution as pore fluids. The geometry of the centrifuge models and setup of the tests were designed according to a full-scale landslide triggering experiment in northern Switzerland (near Ruedlingen village) where 130 m^3 of debris was mobilized after an intense artificial rainfall event, with a maximum failure depth of 1.2 m (Askarinejad et al. 2012*a*; Springman et al. 2009, 2012). The hydromechanical characterization of a silty sand (from the Ruedlingen landslide experiment slope — named hereafter as Ruedlingen soil), having a viscous pore fluid under saturated and partially saturated conditions will be discussed. The results of two centrifuge tests on slopes, with either water or viscous pore fluid, are also presented and compared.

Scaling laws

Scaling laws for infiltration of rainfall (macroscopic scale)

The seepage velocity (v) in a porous medium between two points is proportional to the hydraulic gradient (i) between them, according to Darcy's law, and the constant of this proportionality is called the hydraulic conductivity (k). The seepage velocity is N times higher in the model under Ng centrifugal acceleration, as established by Schofield (1980). Therefore the seepage time will be N^2 times less than that of the prototype, as the seepage length is N times shorter in the model at macroscopic scale. This terminology (macroscopic scale) is used to differentiate the grain scale at which the seepage length is the same as at the prototype scale; hence, the seepage time will be N times less than the prototype (Table 1).

The duration of the rainfall in the $1/N$ geometrically scaled model will also be N^2 times less than that of the prototype. The length equivalent of the total precipitated rainfall in the model will be N times less than that of the prototype, and the rain intensity (which is a ratio between the total rain and the rain duration) will be N times higher in the model. Other scaling laws are discussed in the next section, and summarized in Table 1.

Scaling laws for static liquefaction mechanism (grain scale)

The internal mechanism leading to static liquefaction can be explained by the collapse of saturated voids, which results in local and abrupt increase of the pore pressure (Take et al. 2004). The

Table 1. Scaling laws for rainfall parameters, infiltration process, and static liquefaction (after Askarinejad 2013).

Term [dimension]	Prototype	Model
Length (macroscopic) [L]	1	$1/N$
Seepage velocity (macroscopic and grain scale) [L/T]	1	N
Seepage time (macroscopic) [T]	1	$1/N^2$
Total rain [L]	1	$1/N$
Rain duration [T]	1	$1/N^2$
Rain intensity [L/T]	1	N
Length (grain scale) [L]	1	1
Seepage time (grain scale) [T]	1	$1/N$
Hydraulic gradient (macroscopic) [L/L]	1	N
Hydraulic gradient (grain scale) [L/L]	1	N

known value

locally increased pore pressure reduces the effective stress, and hence the shear strength, and can trigger large movements in the soil mass. Therefore, the main focus is on the grain scale and structure of the soil used in modelling the static liquefaction process. The scaling factor of the length (L) at the grain scale will be $L_p/L_m = 1$ (here and hereafter the subscripts p and m stand for prototype and model, respectively) if a similar structure is reproduced in the model, using the same soil as that in the prototype.

The time scale for gravitational falling (T_{impact}) of a particle at grain scale, calculated based on the assumption of falling with zero initial velocity and at a constant acceleration, would be

$$(1)\qquad T_{\mathrm{impact}} = \sqrt{\frac{2L}{a}}, \quad \frac{L_\mathrm{m}}{L_\mathrm{p}} = 1, \quad \frac{a_\mathrm{m}}{a_\mathrm{p}} = N \Rightarrow \frac{T_{\mathrm{impact,m}}}{T_{\mathrm{impact,p}}} = \sqrt{\frac{1}{N}}$$

where L is the falling height; and a is the acceleration. A particle falling on a water-saturated void results in sudden increase in the pore-fluid pressure. The velocity of dissipation of this excess pore pressure can be calculated using Darcy's law. The flow of liquid from points A to B (Figs. 2b, 2c) will be governed by the hydraulic gradient between these two points and the hydraulic conductivity of the medium along the path. The total hydraulic head (H) at points A and B (denoted by subscripts A and B) in the model and prototype can be calculated using the following equation:

$$(2)\qquad H_{\mathrm{A,p}} = \frac{h_\mathrm{p}\gamma_{\mathrm{f,p}}}{\gamma_\mathrm{f}} + \frac{W_\mathrm{p}}{S\gamma_\mathrm{f}}, \quad H_{\mathrm{B,p}} = 0,$$
$$H_{\mathrm{A,m}} = \frac{h_\mathrm{m}\gamma_{\mathrm{f,m}}}{\gamma_\mathrm{f}} + \frac{W_\mathrm{m}}{S\gamma_\mathrm{f}} = Nh_\mathrm{p} + \frac{NW_\mathrm{p}}{S\gamma_\mathrm{f}} = NH_{\mathrm{A,p}}, \quad H_{\mathrm{B,m}} = 0$$

where H is the total hydraulic head, which is the sum of hydraulic pressure head and elevation head. The elevation heads of points A and B are assumed to be zero. (The height difference between these two points are neglected, as shown in Fig. 2c). W is the weight of the falling particle; h is the height of the saturated pore; γ_f is the unit weight of a reference pore fluid at 1 g, which generally is assumed to be equal to the unit weight of water; and S is the cross-sectional area of the pore.

The hydraulic conductivity and the distance between points A and B (H_{AB} and D_{AB}, respectively) are similar in the model and the prototype, if similar soil structure is reproduced in the model. Therefore, the seepage and the pore-pressure dissipation time will be N times faster in the model compared with the prototype:

$$(3)\qquad V_{\mathrm{A-B}} = k\frac{\Delta H_{\mathrm{AB}}}{D_{\mathrm{AB}}} \Rightarrow V_{\mathrm{A-B,m}} = k\frac{N(\Delta H_{\mathrm{AB,p}})}{D_{\mathrm{AB}}} = N(V_{\mathrm{A-B,p}}) \Rightarrow \frac{T_{\mathrm{dissipation,m}}}{T_{\mathrm{dissipation,p}}} = \frac{1}{N}$$

where $V_{\mathrm{A-B}}$ is the seepage velocity from point A to B; and $T_{\mathrm{dissipation}}$ is the time required for the dissipation of the generated excess pore pressure.

It is concluded that under Ng acceleration, the impact time of a particle is reduced by a factor of $\sqrt{1/N}$, while the localized dissipation of the excess pore-fluid pressure will be N times faster than that in the prototype. This means that the increased acceleration provides a condition in which the excess pore pressure dissipates $\sqrt{N}$ faster than the collapse of the soil structure. Therefore, it is recommended to use a pore fluid $\sqrt{N}$ times more viscous than water to reduce the dissipation time in the model. However, the influences of this pore fluid on the properties of the soil should be investigated.

Ruedlingen soil characteristics

Grain-size distribution

The grain-size distribution of Ruedlingen soil has been determined using the wet sieving method for the grains coarser than 0.063 mm, and sedimentation for the finer fraction. The grain-size distribution curves of the soil from three different depths of a test pit, located 15 m from the border of the test slope in Ruedlingen, are illustrated in Fig. 3. Approximately 60% of the soil is categorized as medium to fine sand, with just over 35% silt and less than 5% clay. The average liquid and plastic limits of the soil from different depths are 27% and 20%, respectively. Accordingly, the average plasticity index of the soil is determined to be 7%. Ruedlingen soil can be categorized as medium- to low-plasticity silty sand (SM), based on the Unified Soil Classification System (ASTM 2006).

Shear strength of soil with water as pore fluid

Casini et al. (2010, 2013) performed several conventional drained and undrained triaxial tests on reconstituted and undisturbed soil samples. The results of these tests suggest a critical stress ratio of $\eta_{\mathrm{CS}} = 1.30$ (stress ratio, $\eta = q/p'$; where $q = \sigma'_\mathrm{a} - \sigma'_\mathrm{r}$; $p' = (\sigma'_\mathrm{a} + 2\sigma'_\mathrm{r})/3$; σ'_a is the effective axial stress; and σ'_r is the effective radial stress applied in the triaxial cell), corresponding to a critical state friction angle of about $\phi'_{\mathrm{CS}} = 32.5°$. Moreover, a series of unsaturated constant shear drained (U-CSD) triaxial tests were performed by Askarinejad (2013) to replicate the stress path that a soil element experiences during the rise of pore-water pressure in a slope, during and (or) after a rainfall event. The results suggest that the stress ratio attained at failure is generally higher than the critical state stress ratio ($\eta_{\mathrm{U\text{-}CSD}} = 1.47$) determined by conventional drained or undrained triaxial tests. Several direct shear tests have also been performed on statically compacted unsaturated specimens, and the results show peak internal friction angle of 30° (Colombo 2009).

Water retention curve

Wetting and drying branches of the water retention curve (WRC) of a natural sample taken from the depth of 0.7 m adjacent to the Ruedlingen test site are shown in Fig. 4. The curve is determined using the axis translation technique (Hilf 1956). The differences between the curves determined by Casini et al. (2010) (Fig. 4) and the results of this research can be attributed to the difference in the initial void ratio of the specimens.

The data of suction versus volumetric water content obtained from both drying and wetting paths have been fitted with two different proposed models from the literature, namely, a unimodal van Genuchten model (van Genuchten 1980) and a bimodal model proposed by Durner (1994).

The unimodal van Genuchten formula is given in eq. (4). The subscripts a and w in this equation stand for air and water, respectively, and $(u_\mathrm{a} - u_\mathrm{w})$ determines the suction in the soil matrix. The denominator (S_{AE}) of the fraction is defined to be the air-entry value of the water retention curve, and λ is the dimensionless fitting parameter.

Fig. 2. (*a*) Schematic of slope subjected to rainfall; (*b*) structure of soil pores showing one particle impacting another one vertically; (*c*) conceptual model under *Ng* acceleration (Askarinejad 2013).

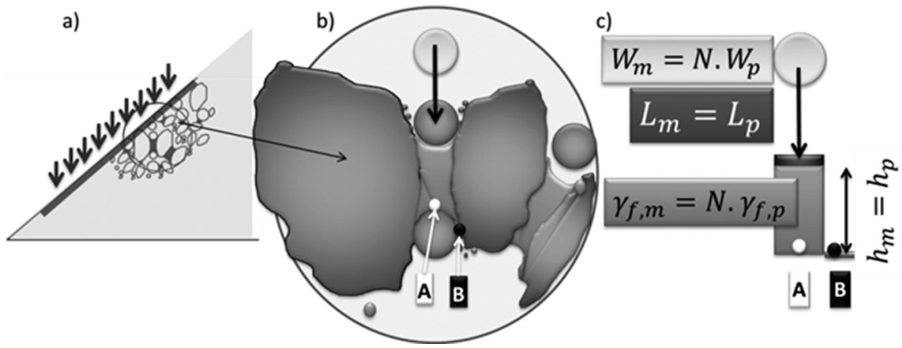


Fig. 3. Grain-size distribution of Ruedlingen soil sampled from different depths (Askarinejad 2013).

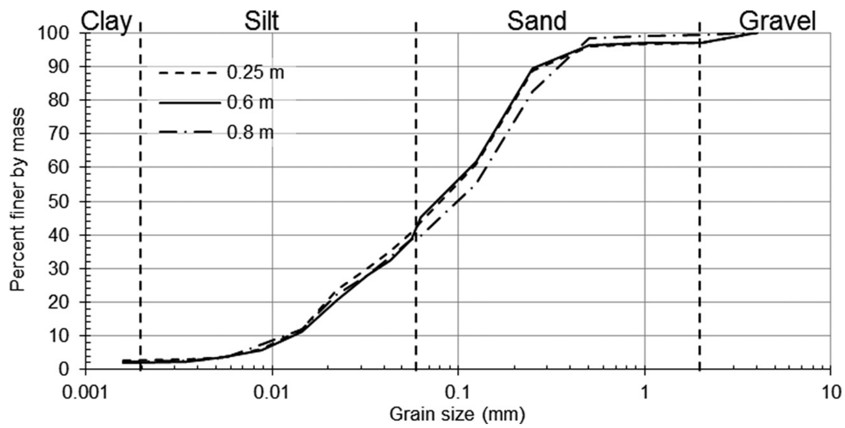
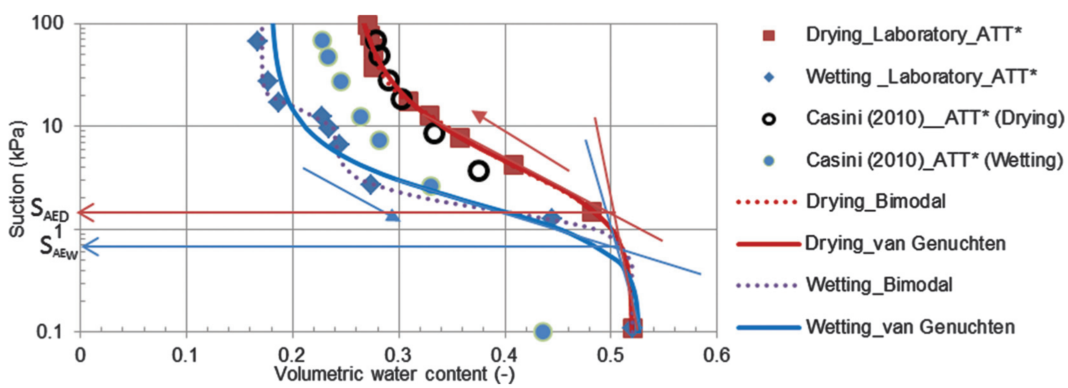


Fig. 4. Water retention curve obtained from natural sample from Ruedlingen test site determined using axis translation technique (ATT) (after Askarinejad 2013). D, drying; S_{AE}, air-entry value; W, wetting.



$$S_e = \left[1 + \left(\frac{u_a - u_w}{S_{AE}}\right)^{1/(1-\lambda)}\right]^{-\lambda} \tag{4}$$

S_e is the relative degree of saturation as defined in eq. (5). The additional subscripts s and r to the saturation degree (S_r) stand for saturated and residual, respectively.

$$S_e = \frac{S_r - S_{r\_r}}{S_{r\_s} - S_{r\_r}} = \frac{\theta - \theta_r}{\theta_s - \theta_r} \tag{5}$$

where θ is the volumetric water content; θ_r is the residual volumetric water content; and θ_s is the saturated volumetric water content.

Table 2. Unimodal van Genuchten parameters for WRC of natural, undisturbed Ruedlingen soil sample (after Askarinejad 2013).

	Parameter				
WRC	S_{AE} (kPa)	λ	θ_r	θ_s	R^2
Wetting branch	0.7	0.537	0.179	0.527	0.971
Drying branch	1.25	0.439	0.252	0.521	0.998

Note: S_{AE}, air-entry value; R^2, regression value; θ_r, residual volumetric water content; θ_s, saturated volumetric water content; λ, dimensionless fitting parameter.

The fitting parameters for these two models were determined using a code written by Seki (2007). The unimodal van Genuchten parameters are listed in Table 2.

Durner (1994) proposed a multimodal water retention curve to describe the retention function of soils having heterogeneous pore

Table 3. Bimodal fitting parameters for WRC of natural, undisturbed Ruedlingen soil sample (after Askarinejad 2013).

WRC	Parameter							
	P_1 (kPa)	λ_1	P_2 (kPa)	λ_2	w_1	θ_r	θ_s	R^2
Wetting branch	1.477	0.7673	14.6	0.8754	0.805	0.171	0.520	0.9993
Drying branch	2.35	0.4815	15.354	0.8056	0.9	0.263	0.520	0.9994

Note: P_1, normalizing pressure for the first part of the curve; P_2, normalizing pressure for the second part of the curve; w_1, weighting factor; λ_1, fitting parameter for the first part of the curve; λ_2, fitting parameter for the second part of the curve.

systems. This method is based on linear superposition of sub-curves of a unimodal van Genuchten type. The general form of a bimodal water retention function is given in eq. (6).

$$S_e = w_1\left[1 + \left(\frac{u_a - u_w}{P_1}\right)^{1/(1-\lambda_1)}\right]^{-\lambda_1} + (1 - w_1)\left[1 + \left(\frac{u_a - u_w}{P_2}\right)^{1/(1-\lambda_2)}\right]^{-\lambda_2} \tag{6}$$

where w_1 is the weighting factor for the first mode of the curve; u_a is the pore-air pressure; u_w is the pore-water pressure; P_1 is the normalizing pressure for the first part of the curve; P_2 is the normalizing pressure for the second part of the curve; λ_1 is the fitting parameter for the first part of the curve; and λ_2 is the fitting parameter for the second part of the curve. The fitting parameters for this model are listed in Table 3.

The results of the bimodal function fit the laboratory data more consistently than the unimodal van Genuchten method (Fig. 4), which can be concluded from the comparison between the regression values (R^2) in Tables 2 and 3. This difference is more pronounced for the wetting branch of the water retention curve.

Durner (1994) suggested that a bimodal water retention curve is associated with a bimodal pore-size distribution, which can be explained using the capillary theory (Burger and Shackelford 2001). The results of mercury intrusion porosimetry (MIP) tests reported by Beck (2011) show that the natural soil samples from the Ruedlingen test site possess two peaks in the pore-size density functions. The radius of the macro pores ranges from 20 to 100 μm, while the size of the micro pores varies from 0.2 to 10 μm.

Viscous fluid

The time scale factors attributed to tests in the centrifuge model and to a prototype regarding excess pore-pressure dissipation and collapse of the soil pores are not the same, as discussed in earlier sections. A possible way of unifying these two time scaling factors is to decrease the hydraulic conductivity of the media by increasing the kinematic viscosity of the pore fluid used in the model by a factor of $\sqrt{N}$. The centrifuge tests performed in this study applied 50*g* acceleration to the models; therefore, the viscosity of the water and glycerine solution used as the pore fluid in these series of tests should be ~7 times more than that of water.

Solution components

Maples (2000) defined the viscosity blending index (VBI) to calculate the viscosity of a blend (Refutas equation):

$$\begin{aligned} &\mathrm{VBI} = 10.975 + 14.535 \ln[\ln(\upsilon + 0.8)] \\ &\mathrm{VBI}_{\mathrm{blend}} = x_{\mathrm{water}}\mathrm{VBI}_{\mathrm{water}} + x_{\mathrm{glycerine}}\mathrm{VBI}_{\mathrm{glycerine}} \end{aligned} \tag{7}$$

where υ is the kinematic viscosity in centistokes (cSt; 1 cSt = 1 mm²/s); and x_{water} and $x_{\mathrm{glycerine}}$ are the weight percentages of water and glycerine in the blend, respectively.

The values of x_{water} = 49.4% and $x_{\mathrm{glycerine}}$ = 50.6% are calculated for a temperature of 20 °C. However, this equation is empirical, and the results are approximate. Therefore, the viscosities of water, glycerine, and viscous solution were tested at different temperatures using a rheometer apparatus. A rheometer measures the shear stress at a given shear rate and calculates the dynamic viscosity of a fluid. The kinematic viscosity is then calculated by dividing the dynamic viscosity by the density of the fluid. The rheometer results showed a weight percentage of x_{water} = 44% and of $x_{\mathrm{glycerine}}$ = 56%. The measured physical properties of the water, glycerine, and viscous solution are summarized in Table 4.

Table 4. Properties of water and viscous solution at 20 °C (after Beck 2011).

Property	Water	Glycerine	Viscous solution (measured)
Dynamic viscosity (cP)	1.08	1410	8.726
Density (g/cm³)	0.998	1.26	1.143
Kinematic viscosity (cSt)	1.082	1119.048	7.635

Note: 1 cP = 0.001 Pa·s; 1 cSt = 1 mm²/s.

Gravitational viscous solution content (VSC) measurement

The boiling point of glycerine is ~290 °C at atmospheric pressure. Therefore, an oven (with maximum temperature of 105 °C) connected to a vacuum was used to decrease the glycerine vapour pressure, which reduces the boiling point of glycerine. The oven was connected with a vacuum pump via a cooling trap filled with ice cubes and water, between the oven and the vacuum pump, to avoid glycerine entering the pump and damaging it.

Effect of viscous solution on hydraulic and mechanical properties of Ruedlingen soil

Proctor compaction curves

The compaction curves derived from the standard and modified Proctor tests for Ruedlingen soil containing water and the viscous solution are shown in Fig. 5. The degree of saturation corresponding to the maximum dry density and optimum pore-fluid content is similar for the standard and modified tests and for water (S_{rw}) and viscous fluid (S_{rvs}). A higher compaction energy level, which was applied during the modified Proctor test, results in a lower optimum VSC and higher maximum dry unit weight. The major effects of viscous solution on the compaction curve are as follows:

(*i*) The change in the dry unit weight of the sample with increasing viscous fluid content is less pronounced, and the curves are flatter compared with the curves of obtained for the soil with water as pore fluid.
(*ii*) The maximum dry unit weight is lower, compared with the soil with water as pore fluid, and occurs at a higher optimum pore-fluid content.

The results illustrate that the compaction efficiency decreases if a more viscous fluid is used. This can be attributed to the fact that the hydraulic conductivity of the soil decreases with increase of the pore-fluid viscosity; therefore, the dissipation of excess pore pressures is slower.

Viscous solution retention curve (VSRC) of Ruedlingen soil

The simultaneous changes in the volumetric viscous solution content and suction are measured in a hydraulic column to derive the VSRC of Ruedlingen soil. The fluid retention curves of Ruedlingen soil, both with water and a viscous solution, are plotted as suction versus degree of saturation in Fig. 6. The air-entry value is

Fig. 5. Comparison of Proctor curves (standard and modified) for viscous pore fluid and water (dashed lines, water; solid lines, viscous solution) (after Beck 2011).

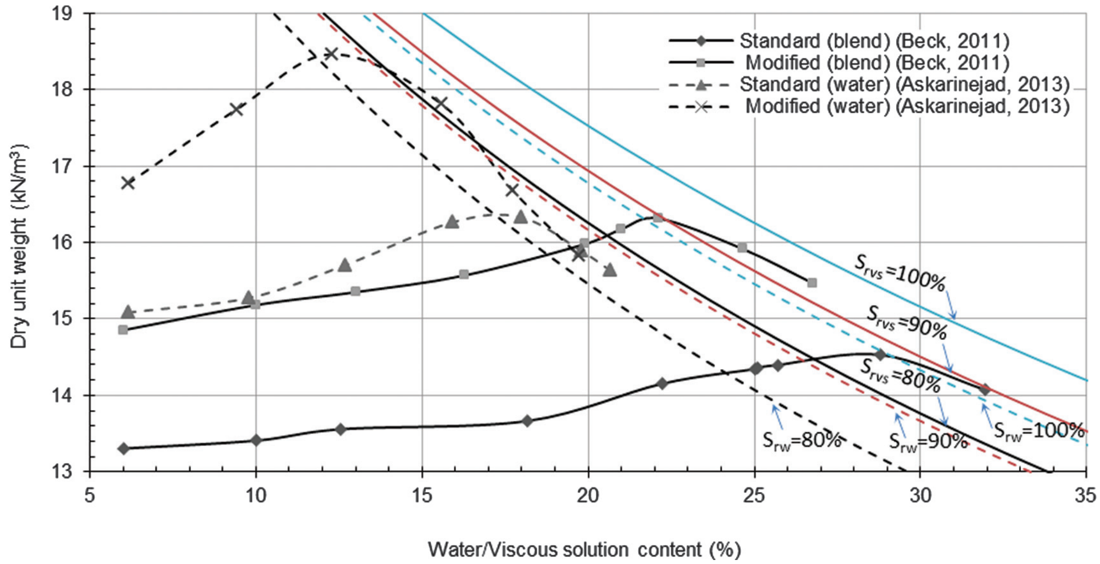
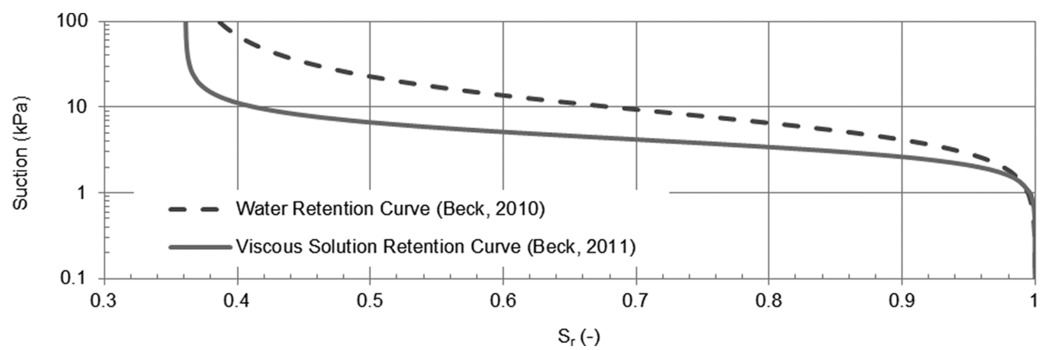


Fig. 6. Comparison of fluid retention curves of Ruedlingen soil with water and viscous solution (after Beck 2011).



slightly higher for the soil with water (~2.5 kPa) compared with the specimen with viscous solution as pore fluid (about 2 kPa). This can be attributed to the lower value of surface tension of the viscous solution compared with water. The air-entry value can be written in terms of surface tension, wetting angle, and corresponding diameter of pores:

$$S_{AE} \propto \frac{4\sigma_f \cos(\theta_f)}{d_{AE}} \quad (8)$$

where S_{AE} is the air-entry value (kPa); σ_f is the surface tension of the wetting fluid, which is 66 mN/m for glycerine and 72 mN/m for water at 20 °C (Bollrich 2007); θ_f is the wetting angle of the wetting fluid, which is ~0° for both glycerine and water (Bollrich 2007); and d_{AE} is the pore diameter for air entry, which is dependent on the soil structure and does not change with the pore fluid.

Hydraulic conductivity function of Ruedlingen soil with viscous solution

The unsaturated hydraulic conductivity function of Ruedlingen soil containing a viscous pore fluid was determined using the instantaneous profile method, as proposed by Daniel (1982). The tensiometers used in these tests, and also in the direct shear tests, were 2100F soil-moisture probes with a maximum suction measurement capacity of 85 kPa. Volumetric fluid contents were derived using time-domain reflectometers (TDR) (EC-5 Decagons), which measure the dielectric constant of the media. The TDRs were calibrated for Ruedlingen soil with pore fluids of water and the viscous solution. The calibration function was fitted to the measurements using a third-order polynomial regression function.

The unsaturated hydraulic conductivity function of Ruedlingen soil, with a viscous solution as the pore fluid, was determined. The changes in suction were required to calculate the hydraulic gradient, and the corresponding changes in the volumetric fluid contents and passing volume of fluid were derived from the viscous solution retention curve (Fig. 6). The result is shown in Fig. 7, where the hydraulic conductivity function of Ruedlingen soil with viscous solution is compared with the conductivity function of the same soil with water.

The functions have a similar shape, and are relatively parallel over the range of suction values, with higher hydraulic conductivity for water. However, the two curves are closer at lower suctions and differ by a factor of less than 10 in the suction range presented.

The values of conductivities for low suctions (s < 3 kPa) could not be determined using the instantaneous profile method because the volumetric fluid content stayed relatively constant at suctions lower than the air-entry value, which resulted in values near zero of calculated passing fluid volume, despite the nonzero hydraulic gradients. Accordingly, the results are shown only for suction values of more than 3 kPa. The maximum values of hydraulic conductivity of the soil with the viscous solution and water are 3.8×10^{-6} and 2.2×10^{-5} m/s, respectively. The saturated conductivity of Ruedlingen

Fig. 7. Comparison between conductivity functions of Ruedlingen soil with water and viscous solution as pore fluid (Askarinejad 2013).

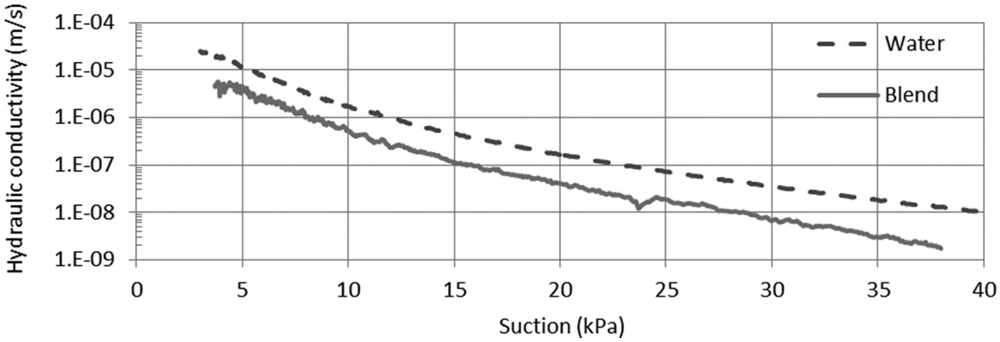
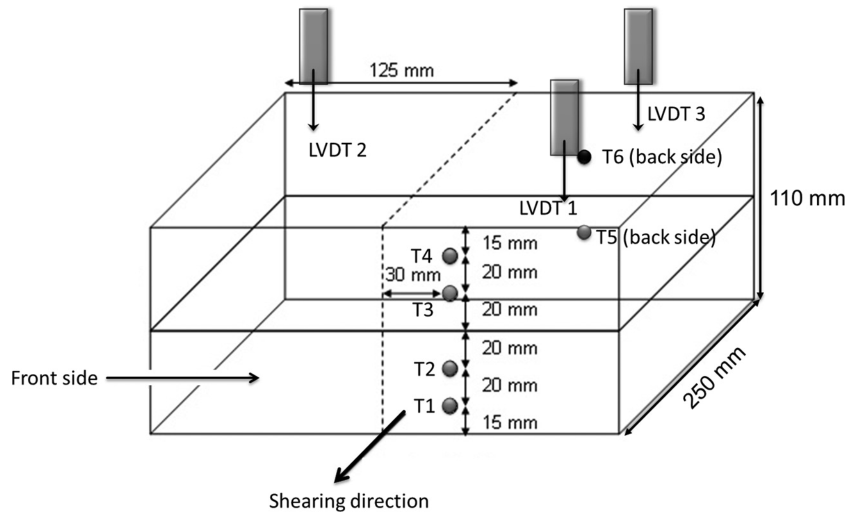


Fig. 8. Diagram of ETHZ–IGT unsaturated direct shear box (Colombo 2009; after Beck 2011). LVDT, linear variable differential transformer.



soil with water is ~6 times greater than that with the viscous solution. However, this ratio should theoretically be 7. The accuracy of the suction measurements, changes in the temperature (which affects the viscosity), and the accuracy of the rheometer apparatus might have caused this difference.

Unsaturated shear strength of Ruedlingen soil with viscous solution

Two series of unsaturated direct shear tests were performed to investigate the effects of the viscous solution as a pore fluid on the shear strength parameters of Ruedlingen soil. Specimens with three different viscous solution contents ($VSC_1 = 15\%$, $VSC_2 = 24\%$, and $VSC_3 = VSC_{sat}$; VSC_{sat}, viscous solution content at saturation) were sheared after consolidation under normal pressures of 8.5, 17, and 33 kPa. Values of suction were measured at four different elevations and at six points (T1–T6), while the VSC was kept constant during shearing in this series of tests. A schematic diagram of the direct shear box is shown in Fig. 8.

A specimen with an initial viscous solution content of 15% was consolidated and sheared in the second series of direct shear tests to a shear stress less than the shear strength of the Ruedlingen soil. This was done to replicate the anisotropic stress condition of a soil element at depth of 1.2 m in the Ruedlingen experiment slope. Details of the anisotropic stress calculations are discussed in Askarinejad (2013). The pore-fluid pressure was increased under constant normal and shear stresses until failure (unsaturated constant shear drained (U-CSD) direct shear test). This process replicated the behaviour of a soil element located at the shear zone in the Ruedlingen slope, under increasing pore-water pressure.

The specimens were prepared in all of the tests using the static compaction method on soils with 15% of VSC. The additional viscous solution required for specimens with higher VSC was added by injection from the base of the shear box, before consolidation, because different initial VSCs might have caused differences in the macro and micro soil structure, as studied by Colombo (2009). The time required for uniform distribution of the additional injected viscous solution was monitored by tensiometer readings (Fig. 8). In addition, the final values of VSC at different depths were measured after each test to check the uniformity of the specimens.

The results of both test series are shown in Fig. 9, in terms of total normal stresses (σ), and in Fig. 10, in terms of effective normal stresses, where τ is the shear stress. The effective stresses (σ') are calculated according to the equation proposed by Bishop (1959):

$$\sigma' = (\sigma - u_a) + \chi(u_a - u_f) \tag{9}$$

where u_a is the pore-air pressure; χ is the effective stress parameter and is assumed to be equal to the degree of saturation ($\chi = S_r$) (as suggested by Jommi (2000), among others); u_f is the pore-liquid pressure, $\sigma - u_a$ is defined as the net stress; and $u_a - u_f$ is the matric suction.

The effective friction angle at large strains was calculated to be between 27.5° and 33°, having a best regression value of 29.5°, based on the results of these tests. This value is quite similar to that derived from the results of the direct shear tests performed on Ruedlingen soil with water as the pore fluid, which was 29.3° (Colombo 2009).

Accordingly, it can be seen from the results of these tests (Fig. 10) that the effective internal friction angle of reconstituted Ruedlingen soil was not affected significantly as a result of using

Fig. 9. Results of unsaturated direct shear tests on Ruedlingen soil with viscous fluid in τ–σ space (Askarinejad 2013).

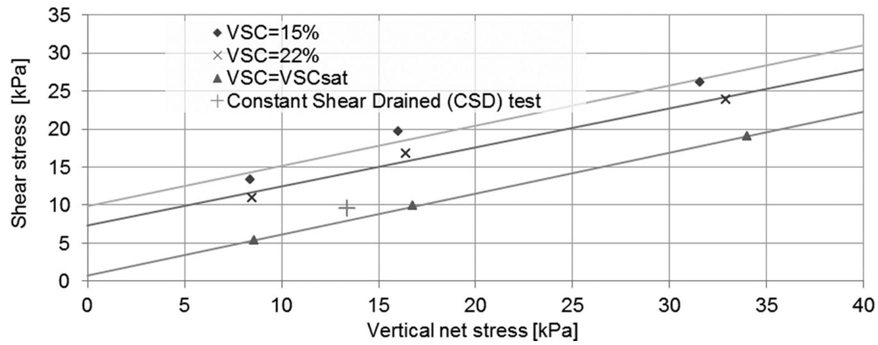


Fig. 10. Results of unsaturated direct shear tests on Ruedlingen soil with viscous fluid, in τ–σ' space (Beck 2011).

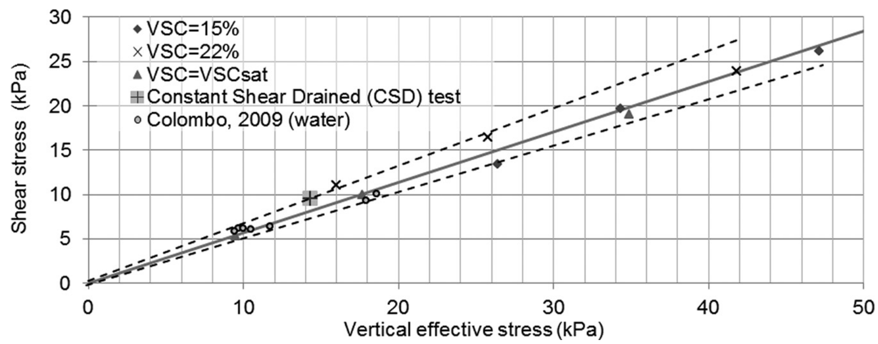
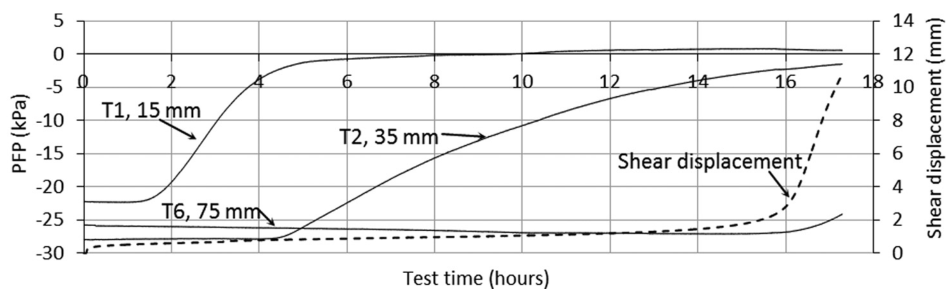


Fig. 11. Changes in pore-fluid pressure (PFP) and shear displacement of specimen under constant shear stresses and increasing pore pressure (Askarinejad 2013). (Time zero is the start of fluid injection from the base.) T, tensiometer; refer to Fig. 8 for location of each tensiometer.



the viscous solution as the pore fluid, although more scatter is observed in the data for these tests. The scatter in the data might be due to the change in temperature, which has an effect on the surface tension of the pore fluid. The precision of the measurements and uniformity of the specimens also affect the consistency of the results.

The changes in pore-fluid pressure (PFP) and shear displacement of the specimen in the second series of tests are shown in Fig. 11. The pore-fluid pressure increases first in the lower parts of the specimen, as the viscous solution is injected into the model from the base of the shear box. The wetting front reaches tensiometer T1 at time 1.44 h, which corresponds to an equivalent hydraulic conductivity of ~10^{-7} m/s, which is slightly higher than the values of the unsaturated hydraulic conductivity measurements at the corresponding suction (Fig. 7). The second tensiometer (T2), installed at a height of 35 mm from the bottom of the specimen and 20 mm below the shear zone, started to react ~4 h after the start of fluid injection; however, the increase of pressure at this point was much slower than that of T1. This can be attributed to the slightly higher values of suction in this region compared with the lower zones, which decreased the hydraulic conductivity of the soil.

The wetting front reached a height of 75 mm (20 mm above the shear surface) at time 16 h. The tensiometer installed at this location (T6) showed a minor increase in suction during the fluid injection, which might be due to the volumetric strains of the specimen, or evaporation of the fluid from the upper part of the box.

The shear displacement (dashed line in Fig. 11) increased slowly at a stable rate until a time of 15 h, when the shear strains accelerated, at which point the pore-fluid pressure at the nearest location to the shear zone (measured by T2; Figs. 8, 11) was –3 kPa, corresponding to a saturation degree of 82%, based on the VSRC (Fig. 6).

The corresponding point of this test (constant shear test) in the τ–σ' space, shown in Fig. 10, lies on the upper band of the internal friction angle, which is ~33°. Similarly, the results of the constant shear drained (CSD) tests under triaxial stress conditions also revealed a higher value of internal friction angle (34°), compared with the conventional drained and undrained triaxial compression tests (Casini et al. 2010, 2013; Askarinejad 2013).

Centrifuge tests

Two tests were performed to study the effects of the pore-fluid viscosity on the responses of slopes subjected to heavy rainfall in the centrifuge. A viscous solution (VS) was used as the pore fluid in the soil, and also as the applied rainfall with intensity of 2.5 mm/h (prototype scale) in the first test (T_VS), while water was used in the second one (T_W) subjected to the same rain intensity in the prototype scale.

Both tests were conducted under 50*g* centrifugal acceleration. The models were prepared in the strongboxes by moist tamping of Ruedlingen soil with 15% gravimetric fluid content, and then trimming to form a 38° slope overlying bedrock parallel to the surface. The top surface of the bedrock was roughened by gluing a thin layer of Ruedlingen soil (Askarinejad et al. 2014). Initial void ratio of the soil was equal to 1.0. This method of model preparation was used to reproduce a soil structure similar to the prototype with a given initial water content and void ratio, although "younger" reconstituted models normally have larger voids compared with the natural samples (Burland 1990).

Two Druck PDCR81 pore-pressure transducers (PPTs) were installed at similar corresponding locations in both models at the interface of the soil and bedrock (P3 and P7 in Fig. 12; PPT3 and PPT7 in Fig. 13). The cross section of the models is also shown in Fig. 12. The surface displacements were measured using inclined particle image velocimetry (iPIV) analysis of the images, captured by photogrammetry cameras at 1 Hz frequency (Askarinejad 2013).

The scaling law for the seepage and infiltration time in these two models is influenced by the hydraulic conductivity of the soil. The seepage time can be calculated using Darcy's law:

$$v = ki \Rightarrow \frac{t_p}{t_m} = \frac{\Delta l_p/(k_p i_p)}{\Delta l_m/(k_m i_m)}$$

$$\xRightarrow{\Delta l_p = N\Delta l_m,\ i_p = (1/N)i_m} \begin{cases} \dfrac{t_p}{t_m} = N^2 \quad (\text{for water: } k_p = k_m) \\ \dfrac{t_p}{t_m} = \dfrac{N^2}{\sqrt{N}} \quad (\text{for viscous solution: } k_p = \sqrt{N}k_m) \end{cases} \tag{10}$$

where v is the seepage–infiltration velocity; k is the hydraulic conductivity; i is the hydraulic gradient; t is the seepage time; Δl is the seepage length (macroscopic); and subscripts p and m stand for prototype and model, respectively. Accordingly, the scaling law for time for the model with the viscous solution is $\sqrt{N = 50} \approx 7$ times less than that of the model with water.

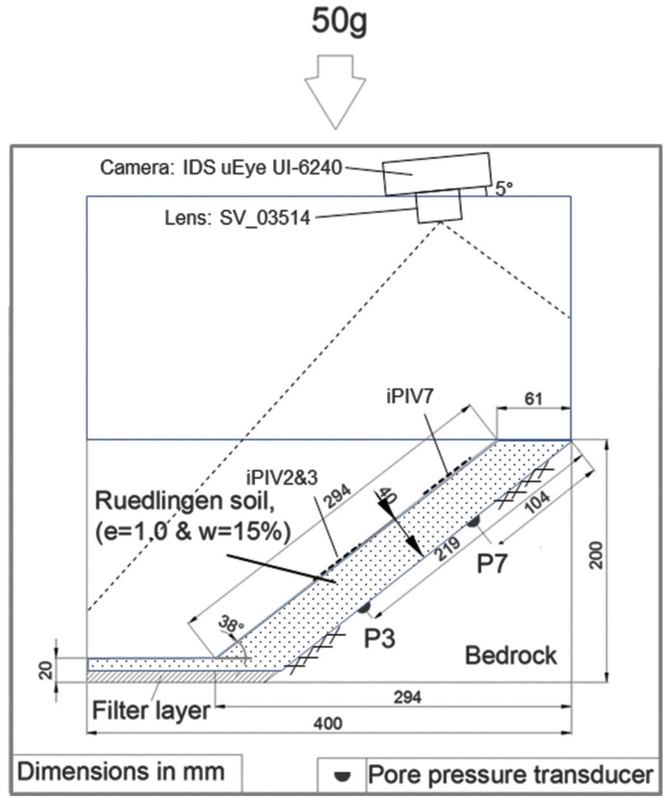
Slope with viscous solution as pore fluid

The sequence of the movements of the slope, having the soil mixed with viscous solution, is shown in Fig. 13. The upper part of the slope started to move 5 h after the start of rainfall, according to the iPIV analysis (Fig. 14). The evolution of the pore-fluid pressures (PFPs) are shown in the Figs. 14 and 15. The first PPT to react was the one at position P7 (PPT7). Pore-fluid pressure at this point started to increase at time 12 h. The pore-pressure rate decreased slightly when the tension cracks appeared at 35 h, and the movements accelerated. The PPT in the compression zone (P3) reacted at time 40 h, and the pore pressure increased at approximately twice the rate of the PFP measured by P7 in the extension zone (Fig. 15). The movements accelerated at time 40 h. Surface erosion started to occur near the left border of the slope (looking downwards) at time 41 h (Fig. 13), and the rain was stopped at time 45 h to prohibit further erosion. However, not only did the movements of the slope not stop, in fact they accelerated, although there was no further rain input and therefore no surface infiltration into the slope. The pore pressures increased until peak values were attained approximately at time 53 h, i.e., 8 h after the rain stopped.

Slope with water as pore fluid

The slope was subjected to rainfall of 2.5 mm/h for 120 h (prototype scale). The slope did not show large surface movements until 34 h after the start of the rainfall, although the iPIV analysis revealed an increasing trend in the velocity of the surface movements from the beginning of the rain. The sequence of the movements of the slope, having the soil mixed with water, is shown in Fig. 16. Tension cracks appeared at the crest 34 h after the start of rainfall (Fig. 16), and the movements accelerated. The first PPT to react was the one at position P7 (Fig. 14). Pore-water pressure at this point started to increase rapidly at 29 h for the following 5 h. The pore-pressure rate decreased slightly when the tension cracks opened at 34 h. The PPT in the compression zone (P3) reacted at time 62 h (Fig. 15), and the pore pressure increased at a similar rate to P7. The movements accelerated at this time, when a sudden increase in the pore-water pressure at the toe of the slope was observed.

Fig. 12. Schematic of centrifuge tests: slope with parallel bedrock (pore fluid: water or viscous solution). *e*, void ratio; *w*, liquid content.



Summary and discussion

The processes involved in static liquefaction can be separated in two parts: (*i*) collapse of the saturated voids due to wetting or contractive behaviour as a result of shearing, which might lead to a local but abrupt increase of the pore-water pressure and (*ii*) dissipation of this sudden excess pore pressure. Each of these processes has its own time factor, when modelled in an increased acceleration field. It was shown that the dissipation of excess pore pressure occurs $\sqrt{N}$ times faster than the collapse of the voids (where *Ng* is the centrifugal acceleration), given that the soil structure reproduced in the model is the same as that in the prototype and the same pore fluid is used in both cases. Therefore, the hydraulic conductivity of the soil should be reduced to compensate the relatively quick dissipation of excess pore pressure. One option to decrease the hydraulic conductivity of the medium is to increase the viscosity of the pore fluid. However, an understanding of the influences of a pore fluid with higher viscosity on the unsaturated and saturated mechanical behaviour of the soil was required.

Fig. 13. Sequence of events on Ruedlingen model slope with viscous solution, captured by camera schematically shown in Fig. 12. Times are at prototype scale (after Askarinejad 2013).

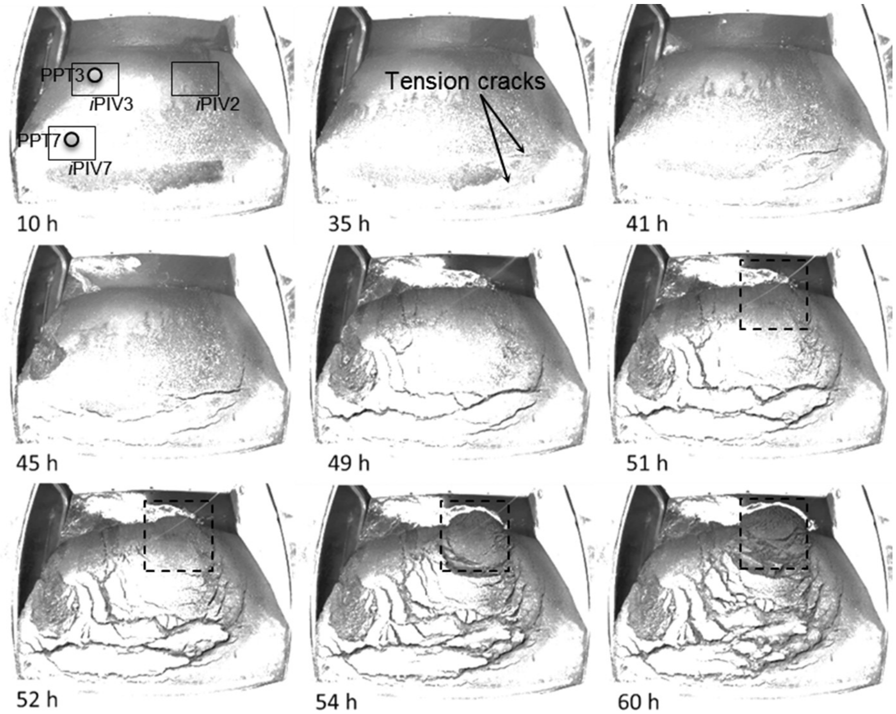
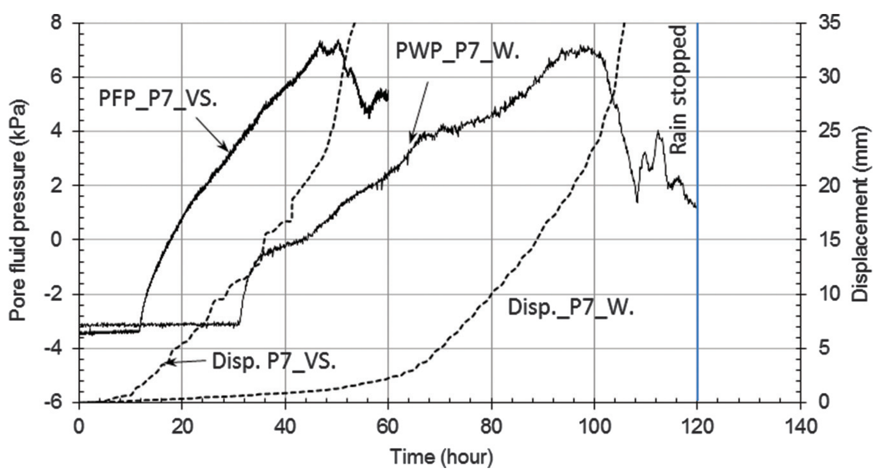


Fig. 14. Pore-water pressure above bedrock at location P7 and surface displacement at same location in tests T_VS and T_W (Disp., displacement; PFP, pore-fluid pressure; PWP, pore-water pressure; VS, viscous solution; W., water) (Askarinejad 2013).



Therefore, the influences of the viscous pore fluid on mechanical and hydraulic characteristics of a silty sand soil were studied. Moreover, a series of centrifuge tests were performed at $N = 50g$. The pore fluids used for the centrifuge modelling were water and a solution of glycerine and water, which provide a higher ($\sqrt{N=50} \approx 7$ times) viscosity compared with water. Therefore, the scaling factors between model and prototype for the dimensionless time factors of collapse of the saturated voids and dissipation of excess pore pressure were similar for the model with viscous solution as the pore fluid.

Effect of viscous pore fluid on properties of soil

- Proctor tests show that more compaction energy is needed to reach the same dry unit weight if a viscous pore fluid is used. Slower excess pore-pressure dissipation due to lower hydraulic conductivity during dynamic compaction is one possible explanation for this observation.
- The hydraulic conductivity function of the soil with viscous solution is determined using the instantaneous profile method (Daniel 1982). This method consists of measuring the suction and viscous solution content profile, in an infiltration column over time during the infiltration process, and retention curves are derived from data with either water or the viscous solution as the pore fluid. The air-entry value was measured to be slightly higher for the soil with water compared with the specimen with viscous solution as pore fluid. This can be due to the lower value of surface tension of the viscous solution compared with water. The hydraulic conductivity function is calculated using the suction measurements and the fluid retention curve and is approximately one order of magnitude lower, when us-

Fig. 15. Pore-water pressure above bedrock at location P3 and surface displacement at same location in tests T_VS and T_W (Askarinejad 2013).

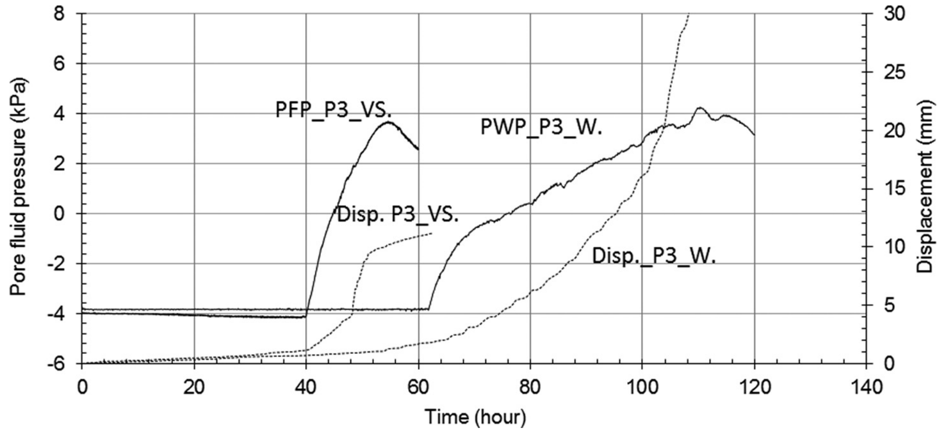
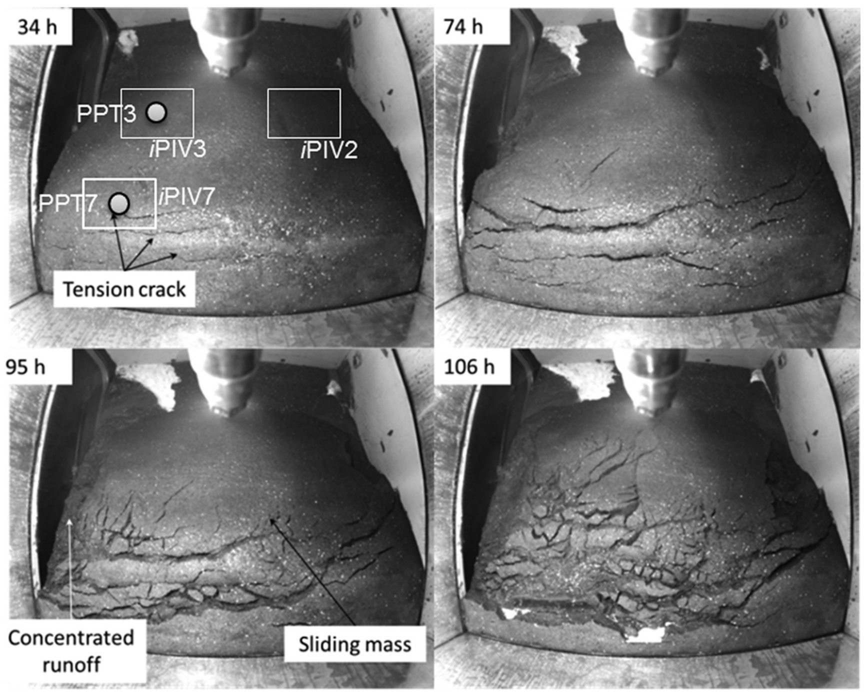


Fig. 16. Sequence of events on Ruedlingen model slope with water. Times are at prototype scale.



ing the viscous solution compared with water, although the ratio is not constant over the whole suction range.

- Direct shear tests were carried out under unsaturated and saturated conditions with the soil mixed with the viscous solution. The shear strength parameters are not significantly affected by the viscous fluid. However, the scatter in the data increases, which can be due to the higher sensitivity of the fluid properties to the changes in the ambient temperature.

Results from centrifuge tests

Initial pore pressure in both models was approximately −4 kPa, and it remained almost constant until the wetting front reached the bedrock surface and the perched water table started to rise. The reaction of the slope with the viscous solution, in terms of acceleration of movements and increase of the pore pressures due to rainfall, occurred sooner than in the model with water. Sudden changes in the rate of movements and pore pressures occurred fairly simultaneously in the model with the viscous solution.

One of the main differences between the behaviour of the two slopes in terms of surface movements is the speed of the movements, which changed significantly in the model with viscous solution after the first increase in the pore pressures. This is dissimilar to the acceleration of the surface movements in the model with water, which followed a smooth continuous curve.

The pore pressure in the model with water increased at a slower rate after the first sudden increase; this drop in the rate of pore pressure increases was more obvious in the measurements made in the extension zone (upper part) of the slope. The pore pressures increased at lower rates until failure occurred in the model. These observations might be attributed to the effect of pore-pressure dissipation rate on the failure mechanism of the slopes triggered by rainfall. It should be noted that the sudden acceleration of movements after the abrupt increase of pore pressures occurred under negative pore-pressure conditions.

Small fluctuations in the PFP were detected at the contraction zone of the slope (PPT3) with viscous solution after the termination of the rainfall (Fig. 17). The PFP measured by PPT3 started to decrease immediately after the rain stopped but then began to increase at a rate slightly higher than that during the rainfall. This cycle repeated several times and finally led into large deformations. A local rotational failure at the right part of the toe of the

Fig. 17. Enlarged view magnifying fluctuation of pore-fluid pressure at interface of soil and bedrock at location P3 after rain was terminated and surface displacement at same location in test T_VS.

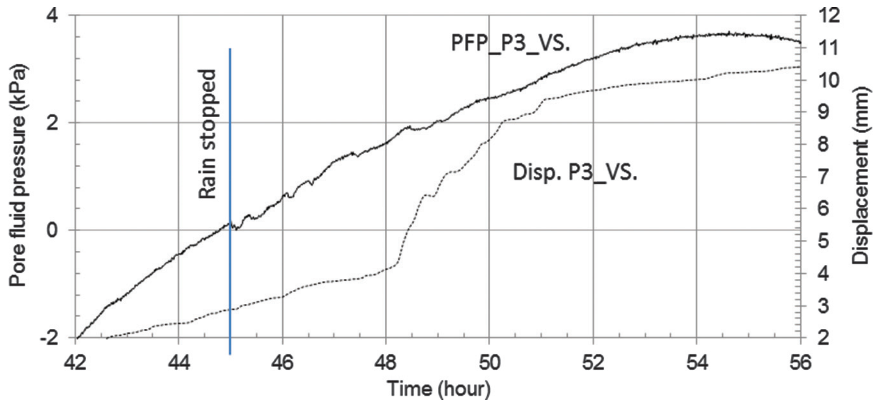
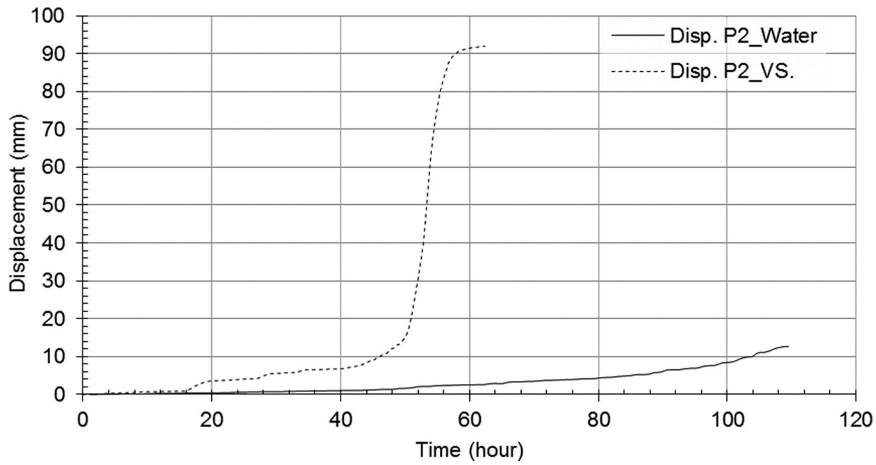


Fig. 18. Comparison between surface movements of slope at location P2 for two slopes with water and viscous solution (location of iPIV2 is shown in Figs. 13 and 16).



slope began to occur at time 51 h (dashed-line box in Fig. 13). This observation was not detected in the model with water. This local failed area at the lower part of the slope with viscous solution was located at the iPIV2 zone (Fig. 13) and travelled longer and faster compared with the corresponding point in the model with water (Fig. 18). The occurrence of this small and local failure at the toe of the slope might be due to a combination of a possible local heterogeneity of the soil and the change in the slope of the underlying impermeable bedrock, which might favour the accumulation of the infiltrated water at this particular location compared with the upper inclined part of the slope. The latter hypothesis has been studied and verified by Take and Beddoe (2014). However, it is not possible to conclude that this observation is an evidence of static liquefaction due to lack of pore-pressure measurements at this location.

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